

Task 4 Write-Up: Optimal Externalization Threshold

Strategy Recap

For a given horizon τ and model M , the strategy externalizes a trade if its predicted adversity probability p exceeds threshold θ . Externalized trades contribute zero PnL. The total PnL is the sum of PnL across all internalized trades only.

Threshold Sweep Results ($\tau=15$, representative)

θ is swept over $[0, 1]$ in steps of 0.1. For each θ , total PnL on the validation set is computed. The θ^* maximising this is selected. Final performance is evaluated on the held-out test set.

Client	$\theta^* (\tau=5)$	$\theta^* (\tau=15)$	$\theta^* (\tau=30)$	Test PnL ($\tau=15$)	Test PnL ($\tau=30$)
A	0.6	0.5	0.6	14,605	21,096
B	0.6	0.5	0.5	11,873	15,082
C	0.6	0.6	0.5	9,661	9,540
D	0.5	0.6	0.3	2,133	0
E	0.5	0.4	0.4	14	3
F	0.6	0.4	0.5	5	-210

Table 5: Optimal thresholds and test PnL by client and selected horizons.

Over-Externalization (θ too low)

When $\theta \rightarrow 0$, almost all trades are externalized. Revenue drops to near zero because even profitable internalized trades (which the LP would earn the spread on) are externalized unnecessarily. At $\theta = 0.0$, the LP earns nothing — 100% externalization means 0% capture of the spread revenue from noise traders (A, B, C). This is visible in the test results for Client E at $\tau=10$ ($\theta^*=0.0$ produces PnL=0).

Under-Externalization (θ too high)

When $\theta \rightarrow 1$, no trades are externalized — the LP internalizes all flow including highly adverse trades from clients E and F. PnL is then driven down by the large negative PnL contributions from those toxic trades, which the model would have correctly flagged at lower θ . For Client F at $\tau=30$, the model can recover PnL only marginally better than full internalization, reflecting the weak model recall.

Global vs. Client-Specific Thresholds

Client-specific thresholds are clearly superior. The evidence:

- θ^* for profitable clients (A, B, C) clusters around 0.5–0.6 — these clients rarely trigger externalization, preserving most of their spread income.
- θ^* for toxic clients (E, F) at longer horizons drops to 0.3–0.4 — the model is used more aggressively to filter out adverse flow, even at the cost of lower fill rates.
- A global θ optimised over all clients would be a compromise that under-protects against E/F and over-externalizes A/B/C — a net loss relative to client-specific tuning.
- The magnitude difference is stark: a single $\theta=0.5$ for Client F at $\tau=30$ yields test PnL=-210, while a client-specific $\theta=0.5$ for F at the same horizon is roughly breakeven. The model's weak recall on F's adversity means even the best threshold provides limited protection.

Final Test Set PnL (summed across clients, at optimal θ^*)

Total test PnL peaks at $\tau=30$ for profitable clients ($A+B+C \approx 45,700$) driven by the growing informational advantage of the LP at longer horizons for noise traders. Clients E and F contribute negligible PnL even at optimal θ , confirming that a quoting-level response (Task 2, δ^*) or a decision not to trade with these clients is the correct commercial response.